

REPORT

ARTIFICIAL INTELLIGENCE (CSA1717)

Assessment Tool – 2

Student Name: Logeshwari D

Register Number: 192524092

Course Code: CSA1717

Course Name: Artificial Intelligence

Assessment Title: Constraint-Based Problem Solving

TABLE OF CONTENTS

NO	CONTENT	PAGE NO
1	Introduction	1
2	Doctor Shift Scheduling Using Backtracking Search	2
3	Robot Path Finding Using Search Algorithm	4
4	Autonomous Rescue Robot	5
5	Conclusion	7
6	References	8

1. INTRODUCTION

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, planning, decision-making, and problem solving. AI uses intelligent agents and search algorithms to solve real-world problems efficiently. This assessment demonstrates three important AI techniques: Backtracking Search, Robot Path Finding, and Uniform Cost Search (UCS). These techniques

are widely used in healthcare scheduling, autonomous robotics, navigation systems, and disaster rescue operations.

2. DOCTOR SHIFT SCHEDULING USING BACKTRACKING SEARCH

Problem Statement

A hospital must assign three doctors (D1, D2, and D3) to three shifts (Morning, Afternoon, and Night). Each doctor must be assigned exactly one shift, and every shift must have only one doctor.

Constraints

- D1 cannot work the Night shift.
- D2 must be assigned before D3.
- D3 cannot work the Morning shift.
- Only one doctor is allowed per shift.

Objective

To assign doctors to shifts while satisfying all constraints using the Backtracking Search algorithm.

Step 1:

Assign D1

- Try **Morning** ✓
- Constraint satisfied
- (D1 cannot work Night).

Assignment: D1 → Morning

Step 2:

Assign D2

- Morning ✗ (already assigned)
- Afternoon ✓

Assignment: D2 → Afternoon

Step 3:

Assign D3

- Morning ✗ (D3 cannot work Morning)
- Afternoon ✗
- (already assigned)
- Night ✓

Assignment: D3 → Night

CONSTRAINT CHECKING

CONSTRAINT	RESULT
D1 cannot work Night	✓ Satisfied
D2 before D3	✓ Afternoon before Night
One doctor per shift	✓ Satisfied
D3 cannot work Morning	✓ Satisfied
One shift for each doctor	✓ Satisfied

FINAL ASSIGNMENT

DOCTOR	SHIFT
D1	Morning
D2	Afternoon
D3	Night

Solution: All constraints are satisfied. Backtracking finds the valid assignment without needing to backtrack further.

3. ROBOT PATH FINDING USING SEARCH ALGORITHM

Problem Statement

A robot must move in a 5×5 grid from the Start (S) position to the Goal (G). The robot can move only up, down, left, and right. Obstacles cannot be crossed, and each movement has a cost of 1. The robot uses the Manhattan Distance heuristic to determine the shortest path.

Objective

To determine the shortest path from Start to Goal while avoiding obstacles.

Solution

Assume the following grid:

```
S □ □ X □  
□ X □ X □  
□ □ □ □ □  
X X □ X □  
□ □ □ □ G
```

S = Start, G = Goal, X = Obstacle

Step 1

Start at S.

Queue = [S]

Step 2

Expand S.

Queue = [(0,1), (1,0)]

Step 3

Expand the next node.

Blocked cells are ignored.

Queue = [(1,0), (0,2)]

Step 4

Continue expanding nodes.

Visit only unvisited safe cells.

Step 5

Eventually Goal (G) is reached.

Optimal Path

$S \rightarrow (0,1) \rightarrow (0,2) \rightarrow (1,2)$

$\rightarrow (2,2) \rightarrow (2,3) \rightarrow (2,4)$

$\rightarrow (3,4) \rightarrow G$

Total Cost

Number of moves = **8**

Cost per move = **1**

Total Cost = 8

Final Answer

The robot successfully reaches the destination by following the shortest path while avoiding obstacles.

4. AUTONOMOUS RESCUE ROBOT

Problem Statement

An autonomous rescue robot is deployed inside a disaster-affected building to locate survivors. The robot can observe only nearby cells, while obstacles and dangerous areas increase movement cost. It must determine the safest and least-cost path using an AI search strategy.

Objective

To find the minimum-cost safe path using **Uniform Cost Search (UCS)**.

Solution

(a) Constraint Analysis

- Robot moves only up, down, left, and right.

- Obstacles block movement.
- Robot observes only nearby cells.
- Every move costs 1.
- Risk zones add extra cost.
- Robot updates its map while moving.

These constraints force the robot to choose the safest and least-cost route.

(b) Solution Using Uniform Cost Search (UCS)

Steps

1. Start from **S**.
2. Add **S** to the priority queue with cost 0.
3. Expand the node with the lowest cost.
4. Observe neighboring cells.
5. Ignore blocked cells.
6. Calculate movement and risk costs.
7. Insert new nodes into the priority queue.
8. Repeat until Goal **G** is reached.
9. Trace back the minimum-cost path.

(c) AI Concepts

Intelligent Agent

The robot senses the environment, makes decisions, and performs actions autonomously.

Rationality

It always chooses the action with the lowest total cost while avoiding risky areas.

Environment Type

- Partially observable
- Dynamic
- Sequential

- Discrete
- Unknown

(d) Justification

Uniform Cost Search is appropriate because:

- It always finds the minimum-cost path.
- It considers both movement and risk costs.
- It avoids expensive routes.
- It guarantees an optimal solution.
- It is suitable for disaster rescue where safety is important.

AI Concepts

Intelligent Agent

- The robot senses the environment and makes autonomous decisions.

Rational Agent

- Chooses the safest path with minimum cost.

Environment Type

- Partially Observable
- Dynamic
- Sequential
- Discrete
- Unknown

Final Answer

Uniform Cost Search finds the optimal path with the least total cost while avoiding unsafe routes.

5. CONCLUSION

This assessment demonstrates the practical use of Artificial Intelligence techniques in solving different categories of problems. Backtracking Search efficiently solves the doctor scheduling problem by satisfying all constraints. Robot Path Finding identifies the shortest path while avoiding obstacles using

informed search. Uniform Cost Search determines the minimum-cost route for an autonomous rescue robot in a disaster environment.

These AI techniques improve decision-making, optimization, automation, and problem solving in real-world applications such as healthcare, robotics, logistics, and intelligent transportation systems. the implemented AI algorithms successfully solve different analytical problems using intelligent search and optimization techniques. Their practical applications make them valuable in healthcare, robotics, logistics, and autonomous systems.

6. REFERENCES

1. Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach*, 4th Edition, Pearson, 2022.
2. Maslej, N., et al. (2024). *Artificial Intelligence Index Report 2024*. Stanford University Human-Centered AI (HAI).
3. Schmidhuber, J. (2022). *Annotated History of Modern AI and Deep Learning*. arXiv.
4. Python Software Foundation. *Python Documentation*.